USDprep

USD scenes made ready for Nuke

User Manual - version 0.9.2 - by Petr Sajner



USDprep opens the heavy USD scene the CG department delivered, lets you pick the one object you need by clicking it, and writes a small file that Nuke reads - geometry, textures and animation. You do not need to know USD to use it.

Scene in the screenshots: ALab, by Animal Logic (open-source production scene).

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1 Quick start

Five steps from a production scene to geometry in your comp.

1. **Open the scene** with *File* > *Open* - the usual Windows dialog, so network drives, DFS shares and a pasted path all work. A USD scene (`.usd, .usda, .usdc, .usdz`) or an Alembic, FBX or OBJ file (`.abc, .fbx, .obj`), which is converted on the way in (see *Alembic, FBX and OBJ files* below). Dropping the file on the window works too. The file is never modified - USDprep only reads it.
2. **Pick the object.** Click it in the 3D view, or click its name in the list on the right. It turns yellow in both places. One click takes the whole object with everything inside it.
3. **Check the name and place** in *Save as*. USDprep suggests the object's name, in the folder you used last.
4. **Press the green Export button.** A line under the button says what was written and how big it is.
5. **In Nuke**, read the `.usdc` file with a *GeoImport* node. Keep the file and the `_textures` folder next to it together.



USDprep after opening a scene: the 3D view on the left, the *Prep for Nuke* panel on the right. Nothing is selected yet.

That is the whole workflow. The rest of this manual explains the choices you can make on the way, what USDprep changes in the file and why, and what to do with an older Nuke.

Alembic, FBX and OBJ files

An Alembic file (`.abc`) - a camera track, a point cache, an animated character -, an `.fbx` - a model or a character from Maya, 3ds Max, Blender, Unreal or an online store - and an `.obj` - a lidar or photogrammetry scan - are converted to a temporary USD file when you open them; a progress window shows how far it got. From then on it is a scene like any other: pick objects, reduce them, export them. The original file is only read, never changed. The panel header says *converted*; hover over it for what the conversion found and what it had to leave out.

Alembic. Every mesh comes with its animation sample by sample: points that deform, objects that move, cameras with their focal length and film back (in millimetres, a lens squeeze included). UVs, normals, vertex colours and face sets come along, subdivision surfaces keep their creases, and the frames stay the ones the file has. Alembic carries no materials, so the objects come in their vertex colour or grey. Points and curves

come along too, but Nuke's USD reader draws neither. Files written by any application since 2013 (Ogawa) are read; an older HDF5 file is named as such.

FBX. Meshes with their UVs, normals, vertex colours and materials with their textures - also the textures packed inside the `.fbx` - cameras, lights, and the animation of the first take on the frames the file has (a shot that starts at 1001 stays at 1001). A character moved by a skeleton or by blend shapes is baked into a point cache, frame by frame: in Nuke it simply plays, there is no rig to set up. A texture path written on another computer is looked for next to the `.fbx`, in its `.fbm` folder and in the folders around it; a texture that cannot be found is named. An FBX camera keeps its exact field of view; the film back is not in what the file gives, so a 36 mm wide one is assumed and the focal length set to match.

OBJ. Every object or group of the OBJ becomes an object in the list, with its polygons as they are, its UVs, normals and vertex colours, and the materials of its `.mtl` file with their textures (colour, opacity, roughness, metallic, emission and normal maps). A texture path written on another computer is looked for next to the `.mtl`.

A scan at survey coordinates - millions of units from the origin - keeps its exact place in the exported file. The 3D view, which cannot draw that far out precisely, shows it at the origin instead and says *shown at the origin*.

To turn a heavy scan into a light stand-in for the comp, use the **Geometry** slider under *Advanced*: *1/25* and *1/100* of the polygons are there for exactly this. A two-million-polygon scan reduced to a hundredth takes a few seconds and keeps its texture.

The conversion is kept while the program runs, so opening the same file again is immediate; it is cleaned up the next time USDprep starts.

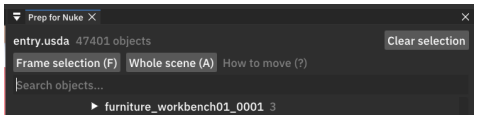
USDprep and usdtweak

USDprep is built on top of **usdtweak**, an open-source USD editor by Cyril Pichard, and it is made for one job: getting USD scenes into Nuke. It always starts the same way - the 3D view and the *Prep for Nuke* panel, nothing else, whatever was open the last time.

The editor underneath has not been removed and it works: its panels (scene outliner, property and layer editors, timeline, text editor) are in the **Windows** menu and stay open until you close the program. USDprep does not build on them and this manual does not cover them. If a USD editor is what you need, install usdtweak itself.

The version you are running is shown in the window title, at the right end of the panel's second row, and in **Help > About**.

2 Finding your way around



The top of the panel shows the scene's name and how many objects it has. **Frame selection (F)** moves the camera to the selected object. **Whole scene (A)** is the way back from anywhere: it shows where the objects are. **Clear selection** deselects everything.

Hover over *How to move (?)* for the mouse controls; they are also listed below.

Moving in the 3D view

Do this	To
Mouse wheel	Move closer or further
Alt + left drag	Turn around the object
Alt + middle drag	Slide sideways, up and down
Alt + right drag	Move closer or further, smoothly
Hold the right button + W A S D (Q / E = down / up)	Walk through the scene; Shift walks faster
F	Frame the selected object
A	Whole scene - where the objects are

Seeing inside rooms. When you frame an object that stands in a room, the wall between the camera and the object is cut away so that you can see it. The walls come back when you press A or walk with the right mouse button. Nothing is removed from the scene - it is only the view.

The object list



The list shows the scene as folders within folders. Click the small triangle to open a folder; click a **name** to select that object with everything inside it. Click the name again to deselect.

The selected object is **yellow**; what belongs to it is a paler yellow. The number after a name says how many parts the object has.

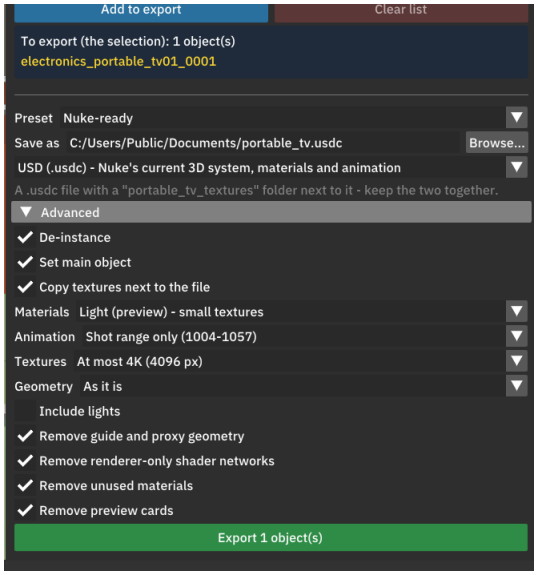
List and 3D view follow each other: click an object in 3D and the list opens exactly the folders needed and scrolls it into the middle.

Up arrow selects the parent - the bigger object this one belongs to. **Down arrow** goes back down. Use them when a click picked a smaller or bigger piece than you wanted.

Ctrl + click adds an object to the selection, or takes one out, without touching the rest.

Search objects... filters the list by name while you type.

3 Exporting



What gets exported. Normally: whatever is selected. If you need several objects from different places, select the first, press **Add to export**, select the next, add it, and so on. The list keeps them; clicking an entry shows you that object again. **Clear list** empties it. All of them go into one file.

Preset. *Nuke-ready* is the one to use: it makes the file small and makes sure Nuke can read everything in it. *Original (nothing changed)* only cuts the object out and leaves it exactly as the CG department made it - Nuke may not show parts of such a file correctly. *Recipe file...* loads settings your pipeline TD prepared.

Save as. Type a name or use *Browse...*. The file extension follows the format below it.

Format. See the table on this page.

Advanced holds every individual switch; chapter 5 explains them. You rarely need it.

While it exports, a bar takes the place of the button: how far it is, what it is doing and for how long. The program stays usable. **Cancel** stops the export and nothing is written; a step that cannot be interrupted (reading a big set) finishes first.

After the export the line under the button reports the result. **Details** opens the full report: everything USDprep did to the file, in plain words. If something in Nuke looks different from what you expected, the reason is in that report.

Which format for which Nuke

Format	Use it for	What is inside	What you get
USD (.usdc)	Nuke with the current, USD-based 3D system (GeoImport, ScanlineRender2). Checked with Nuke 16.1 and 17.0.	Geometry, materials with their textures, animation, cameras.	`name.usdc` `name_textures/`

Alembic (.abc)	An older Nuke that only has the classic 3D system (ReadGeo). Animated objects.	Geometry with UVs and its animation. No materials - Nuke reads none from this format.	`name.abc`
			`name_textures/`
			`name_abc.nk`

OBJ (.obj)	An older Nuke, objects that do not move.	Geometry with UVs at one frame. No animation, no materials.	`name.obj`
			`name_textures/`
			`name_obj.nk`

USDprep offers only formats that Nuke was measured to read. A `.usdz` package is not among them: Nuke loads its geometry but none of the textures inside it.

4 Opening the result in Nuke

USD - the current 3D system

1. Create a **GeoImport** node and point it at the `.usdc` file.
2. Connect it to a **ScanlineRender2** together with your camera. Materials and textures are already in the file.
3. Without any light in the Nuke scene the object shows its plain texture colours, which is usually what a comp needs. As soon as you add a light, it is lit by that light.

Keep the `.usdc` and its `_textures` folder together when you move or copy them - the file points at the textures next to it.

Alembic and OBJ - the classic 3D system

Nuke reads no materials from these two formats, so USDprep writes a small Nuke script next to the file that sets everything up for you:

1. In Nuke choose **File > Insert Comp Nodes...** and pick `name_abc.nk` (or `name_obj.nk`).
2. You get one **ReadGeo** node per material, each with its texture already connected, and a **Scene** node that joins them. Connect the Scene to your ScanlineRender.
3. For an `.abc`, set the Nuke project to the frame rate of the scene; the export report tells you which one it is. The animation sits on the scene's own frame numbers (for example 1004-1057).

When an object has several materials, the geometry is also written once per material into a `name_abc_parts` (or `name_obj_parts`) folder - a classic ReadGeo takes only one texture for everything it reads, and the script uses those files. The paths inside the script are absolute: if you move the export to another place, export again or fix the paths in the Read and ReadGeo nodes.

If you load an `.abc` by hand instead, Nuke asks which items to load - choose all of them.

5 The Advanced switches

Every change USDprep makes to the file has its own switch here, so you are always one click away from the original. The preset sets them; you can change any of them for one export.

Switch	What it does	Nuke-ready
De-instance	Production scenes reuse one object many times (instancing). This turns the copies into ordinary objects. Nuke reads instancing, but its render stops with "Too many open files" on a scene with over a thousand instances. The file does not get larger.	on
Set main object	Marks the main object of the file, which applications use to know what to load.	on
Copy textures next to the file	Copies every texture the object uses into the `_textures` folder and points the file at the copies. Off: the file keeps pointing into the production folders.	on
Materials	Production objects often carry two materials: a heavy one for final renders and a light one for previews. *Light* keeps the small one, which is what a comp needs; *Full quality* keeps the heavy one; *Keep both* leaves them as they are.	Light
Animation	*Shot range only* drops animation from before and after the shot (simulations often start a long time before it). *One frame* freezes the object at the frame you type. *Everything* keeps every sample.	Shot range
Textures	Scales textures larger than the chosen size down - in the exported copy only, the originals are never touched.	At most 4K
Geometry	A slider with six stops: as it is, 1/2, 1/4, 1/10, 1/25 and 1/100 of the polygons. Reduces dense objects, keeping UVs and normals as well as it can; 1/25 and 1/100 are for scans. It changes the shape and is never on by itself. Objects under 500 polygons stay as they are.	As it is
Include lights	Off by default: with a light in the file Nuke stops showing objects in their plain colours and the picture goes dark. On: the light types Nuke can use come along; the others become axes of the same name in the same place, so you can rebuild them.	off
Remove guide and proxy geometry	Stand-in geometry the asset carries for other programs. Nuke draws all of it, so the object would show twice. Off: it stays in the file, hidden.	on
Remove renderer-only shader networks	Material parts only Arnold, RenderMan and similar renderers understand, with the textures only they use.	on
Remove unused materials	Materials nothing in the export uses.	on
Remove preview cards	Six small pictures some viewers show instead of the object. Nuke never draws them.	on

6 What USDprep changes, and why

Nuke does not read everything a USD file can contain. What Nuke 16.1 and 17.0 actually read was measured with test scenes, and the *Nuke-ready* preset converts what they cannot read into the closest thing they can. Nothing is thrown away silently: every conversion is listed, by name, in the report behind the **Details** button.

In the scene	What Nuke would do	What USDprep does
Textures split into UDIM tiles	Black material	Joins the tiles into one texture and adjusts the material to use it
Lights Nuke cannot use (rectangular, cylinder...)	Ignores them	Replaces each by an axis of the same name in the same place (only when *Include lights* is on)
Guide and proxy geometry	Draws it on top of the real object	Removes it, or hides it if you keep it
MaterialX next to a standard material	Black material	Removes the MaterialX part
A material only a production renderer understands	The object disappears	Detaches the object from it; it shows in its plain display colour
Characters moved by a skeleton	Stands frozen in its rest pose	Bakes the motion into the geometry
A scene built with Z pointing up	Lies on its back	Stands it up
One object with several materials on different faces	Grey object	Splits it into one object per material, inside a group with the original name
Basic shapes (sphere, cube, cylinder, cone, capsule)	Draws nothing	Turns them into ordinary geometry
Shape animation (blend shapes, for example a face)	Stays in its base shape	Bakes it into the geometry
8-bit roughness, metallic and normal maps	Reads the numbers wrong: surfaces too glossy, bumps bent	Writes corrected copies of those textures
Curves (hair, whiskers, wires) and volumes (smoke, clouds)	Draws nothing	Cannot be converted - the report tells you they are there

7 If something looks wrong

What you see	Likely reason and what to do
The object is black in Nuke	The textures are not where the file expects them: keep the `_textures` folder next to the file. If you exported with *Original*, export again with *Nuke-ready*.
The object is dark	There is a light in the file or in your Nuke scene. Export without *Include lights*, or light the object in Nuke.
A part is missing	Open **Details** after the export: the report lists what was removed or could not be converted (for example curves). Also check that the part was inside your selection - press the up arrow to select the bigger object.
The report says textures were left out	Those texture files are not on this computer; the scene points at a place you cannot reach. Ask for the textures.

What you see	Likely reason and what to do
An FBX or OBJ object is grey or plain after opening	Hover over *converted* in the panel header: it names the textures that were not found. Put them next to the file, or in a *textures* folder beside it, and open the file again.
Alembic in Nuke does not move, or moves at the wrong speed	Set the Nuke project's frame rate to the one named in the report, and look at the scene's frame numbers.
Only one piece of an .abc shows	You read it by hand and picked one item. Use the `.nk` script, or select all items in the dialog.
The 3D view stays empty or grey	The graphics driver is too old (OpenGL 4.5 is needed). Update the driver from NVIDIA, AMD or Intel. A computer that runs Nuke 16 or 17 is new enough.
I am lost in the 3D view	Press **A**.
A file on a network drive does not open	USDprep opens files through the Windows dialog: if Windows Explorer can open the folder, USDprep can. When a file cannot be read, a window says why.
An exported object has no material	Export again with this version: the materials an object uses now come along even when they live elsewhere in the scene.

8 For pipeline TDs: the command line

Everything the panel does is also available without a window, for scripts and render farms. `usdcut` is installed next to the program (`bin\usdcut.exe`); tick *Add the usdcut command line tool to my PATH* in the installer to call it from anywhere.

```
usdcut extract scene.usd /World/Set/Car -o car.usdc --preset nuke
usdcut extract scene.usd /World/Char/Hero -o hero.abc --preset nuke --frames 1001-1080
usdcut extract scene.usd /World/Set/Car -o car.obj --preset nuke --frame 1010
usdcut select scene.usd --name "*door*" --topmost
usdcut extract scan.obj /scan -o scan_small.usdc --preset nuke --simplify 0.01
usdcut extract character.fbx /character -o character.usdc --preset nuke
usdcut inspect scene.usd     (also an .abc, .fbx or .obj: what the conversion makes of it)
usdcut presets nuke        (prints the preset as JSON - the start of your own recipe)
usdcut help
```

A recipe is a small JSON file with the same switches as the Advanced section. Start from the printed preset, change the lines you need, and give it to artists to load with *Recipe file...* or pass it as `--recipe studio.json`. Add `--report out.json` to get the export report as a file.

9 Installation

Run `USDprep-0.9.2-setup.exe` and follow its pages. That is all.

- It installs for **you only**: no administrator rights and no internet connection are needed.
- Everything the program needs is inside its own folder (by default `%LOCALAPPDATA%\Programs\USDprep`). It does not touch Nuke, Python or any other USD installation on the computer.
- It adds a **USDprep** entry to the Start menu. A desktop shortcut and the `usdcut` entry in your PATH are optional tick boxes.
- This manual is installed with the program; the Start menu has a shortcut to it.
- **To remove it**, use *Settings* > *Apps* or *Uninstall USDprep* in the Start menu. Nothing is left behind.

What the computer needs

- Windows 10 or 11, 64-bit.
- A graphics card with a current driver from its vendor (OpenGL 4.5). Any computer that runs Nuke 16 or 17 qualifies. Without it the 3D view stays empty; `usdcut` works regardless.
- About 100 MB of disk space.

Licence

USDprep was written by **Petr Sajner** (Copyright 2026 Petr Sajner) and is free software under the Apache License 2.0 - free to use, change and redistribute, also commercially. It is built on usdtweak by Cyril Pichard (Apache 2.0), Pixar's OpenUSD, Alembic, Imath and meshoptimizer. Their licences are in the `licenses` folder of the installation, listed in `THIRD_PARTY_NOTICES.txt`.